The DNRM Protocol: A Decentralized Standard for Visual Privacy

The expansion of ubiquitous cameras, AI-enabled surveillance, and automated facial recognition has created a structural imbalance between image capture and individual consent. Existing privacy protections are largely post-capture, platform-dependent, and easily bypassed. This paper proposes the Do Not Record Me (DNRM) Protocol, a decentralized, edge-computed system for automated visual consent enforcement triggered by machine-readable physical markers. When a DNRM marker is detected, facial imagery is obfuscated in real time before storage or transmission, while the original biometric region is encrypted locally and locked behind a Decentralized Recovery and Escrow Network (DREN) governed by blockchain-based consensus, threshold cryptography, and public auditability.

The protocol is designed to solve a hard real-world problem: how to enable privacy by default while still allowing narrowly scoped, lawful recovery of obscured facial evidence in cases involving serious criminal activity. To achieve this, DNRM combines local inference, encrypted biometric compartmentalization, threshold key sharding, immutable warrant logging, and decentralized approval logic such that no single company, government, platform, or device owner can unilaterally recover protected biometric data. This architecture aims to make mass surveillance, silent bulk decryption, and centralized biometric abuse computationally and institutionally impractical, while preserving a viable path for targeted lawful access under transparent, multi-party control.

DNRM is intended as open-source infrastructure for a future in which visual consent is machine-readable, enforceable, auditable, and compatible with decentralized governance.

1. Introduction

Modern society has normalized continuous image capture. Smartphones, smart glasses, action cameras, CCTV systems, drones, social media live streams, and body-worn devices collectively create an environment in which individuals are recorded by default, often without notice and almost never with meaningful consent. In parallel, facial recognition systems, behavioral tracking engines, biometric indexing pipelines, and generative AI training datasets have expanded the downstream value of every captured face.

The central defect in the current paradigm is that visual consent is assumed rather than negotiated. A person can object verbally, but there is no globally recognized, machine-readable, technically enforceable way to express non-consent to being recorded, scraped, profiled, or distributed. Existing controls operate too late in the chain. They depend on the goodwill of platforms, the responsiveness of moderators, the reliability of takedown systems, and the legal burden of the victim.

The DNRM Protocol proposes a different model: consent enforcement at the point of capture.

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DNRM creates a physical-digital bridge in which a person wearing standardized, high-contrast visual markers broadcasts a persistent, machine-readable privacy signal. Compatible capture devices detect that marker locally and automatically obfuscate the wearer's face in real time before the media is stored, shared, or uploaded. At the same time, the original protected biometric region is encrypted and cryptographically sealed such that it cannot be viewed by the device owner, platform, or manufacturer.

However, a privacy system that permanently destroys all recoverability faces immediate resistance from regulators, courts, OEMs, and law enforcement, especially where violent crime, child exploitation, terrorism, trafficking, or other serious offenses may be captured incidentally. A protocol that shields all imagery from lawful review is unlikely to be adopted at firmware or platform level. For that reason, DNRM introduces a second layer: a decentralized blockchain-governed escrow and recovery mechanism that permits tightly scoped decryption only after threshold approval by independent network actors under public, immutable audit conditions.

The result is a hybrid model:

- Privacy by default
- Local obfuscation first
- No unilateral decryption
- Publicly auditable recovery events
- No central biometric honeypot
- No mass recovery pathway

This paper defines the technical, cryptographic, and governance foundations of that model.

2. Design Goals

The DNRM Protocol is built around the following design goals:

2.1 Privacy by Default If a valid DNRM marker is detected, the system must protect the subject automatically without requiring manual action, network connectivity, or platform cooperation.

2.2 Edge-First Enforcement All detection, localization, and obfuscation must occur locally on the capture device before cloud upload or long-term storage.

2.3 Recoverability Without Centralized Control If lawful recovery is permitted, it must occur only through decentralized threshold consensus. No manufacturer, platform, government, or recorder may possess unilateral access.

2.4 No Biometric Honeypot The system must never create a centralized repository of raw facial data, decrypted footage, or reusable master keys.

2.5 Public Accountability Every recovery request, approval decision, rejection, and key release event must be immutably logged to a public ledger or verifiable blockchain layer.

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2.6 Targeted, Not Bulk, Access The architecture must support narrow recovery of specific protected biometric segments relevant to a defined legal request, while structurally resisting mass decryption campaigns.

2.7 Realistic Adoption Path The protocol must be politically and commercially legible to device manufacturers, platforms, regulators, privacy advocates, and courts.

2.8 Open Governance The protocol must evolve through transparent open-source governance rather than proprietary control.

3. System Overview

The DNRM architecture consists of five major layers:

Layer 1: Physical Consent Signaling Machine-readable DNRM markers embedded in apparel, accessories, patches, or visual tags.

Layer 2: Edge Detection and Obfuscation Local computer vision pipeline detects the marker and applies real-time biometric masking before storage or upload.

Layer 3: Encrypted Biometric Compartment The original protected facial region is isolated, encrypted locally, and sealed into a recoverable evidence envelope.

Layer 4: Decentralized Recovery and Escrow Network (DREN) A blockchain-governed, threshold-cryptographic network stores sharded authority over recovery keys, not the footage itself.

Layer 5: Lawful Recovery Interface A narrowly scoped recovery process requiring signed legal authorization, multi-node approval, smart-contract execution, and public audit logging.

This separation is fundamental. The device holds the encrypted compartment. The network governs access to the recovery key. No single side is sufficient on its own.

4. Trust and Compliance Tiers

DNRM must explicitly acknowledge that protection quality depends on the capture environment. The protocol therefore defines three operational trust tiers.

4.1 Tier I: Native Compliant Capture This is the ideal environment. The capture device includes native DNRM support at firmware, operating system, camera pipeline, or application level. Characteristics:

- marker detection at capture time
- real-time obfuscation before file encoding
- encrypted biometric compartment creation
- blockchain-compatible recovery envelope

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- public audit support
- strongest privacy guarantees

Examples:

- smartphone OEM integration
- smart glasses with DNRM firmware
- bodycam systems with certified DNRM support
- drone firmware with native protocol enforcement

4.2 Tier II: Software-Compliant Capture The device does not have deep firmware integration, but an application layer implements DNRM detection and masking. Characteristics:

- protection still possible
- weaker guarantees around pre-buffer raw access
- greater risk of bypass by malicious apps or device owners
- recovery envelope may be app-managed rather than hardware-backed

Examples:

- social apps
- livestream apps
- video editors
- browser-based camera tools

4.3 Tier III: Non-Compliant or Adversarial Capture No DNRM integration exists, or the recorder intentionally bypasses protection. Characteristics:

- no guaranteed protection
- no escrow guarantee
- no reliable obfuscation enforcement
- DNRM marker may still signal non-consent socially or legally, but not technically

Examples:

- second-camera recording
- raw sensor extraction
- modified firmware
- stripped app builds
- hidden surveillance devices

This tiered model is important for credibility. DNRM is not magic. It is a protocol for compliant ecosystems and future standards, not a claim of universal invulnerability.

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5. Phase I: Edge-Computed Detection and Obfuscation Pipeline

The first enforcement stage occurs entirely on-device.

5.1 Marker Design Requirements DNRM markers must be visually distinct, machine-readable, robust under real-world conditions, and resistant to accidental false positives. Core requirements:

- high contrast
- shape consistency
- scale invariance
- partial occlusion tolerance
- resilience under motion blur
- cross-garment adaptability
- minimal aesthetic ambiguity
- optional cryptographic pattern encoding for future signed markers

Markers may exist in multiple approved classes:

- standard apparel prints
- embroidery markers
- patches
- caps
- wearable tags
- sticker-based signals
- reflective variants for low-light environments

Future protocol versions may include:

- signed fiducials
- rotating pattern generations
- anti-counterfeit microstructures
- context-class markers indicating different privacy preferences

5.2 Detection Model Architecture The DNRM detector should prioritize speed, low power, and hardware portability over maximal model size. Suitable architecture families include:

- YOLO nano class detectors
- MobileNet-based detectors
- EfficientDet-lite variants
- specialized quantized CNNs
- NPU-optimized detection stacks

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The objective is not general object understanding, but highly optimized detection of a narrow visual vocabulary. This permits aggressive model compression and hardware acceleration. Required conditions for production-grade deployment:

- low-latency inference
- robust detection in dynamic lighting
- motion blur resilience
- tolerance to body rotation and partial garment folding
- resistance to spoofing with casual lookalikes
- high recall where wearer protection is preferred
- tunable confidence thresholds based on device policy

5.3 Face Association Logic Marker detection alone is insufficient. The system must infer which face or faces correspond to the marker. This requires a face association layer that may combine:

- marker-body spatial relationship estimation
- pose and skeleton tracking
- temporal bounding-box persistence
- multi-object association across frames
- proximity and occlusion reasoning

For example, if a chest-worn marker is detected, the face bounding region will likely sit above the marker within a constrained geometric zone. For a cap marker, the face may lie immediately below. In crowded scenes, association confidence should be tracked over time rather than on a single frame.

Environmental False Positives & Anti-Spam Logic Marker detection alone is insufficient; it must be coupled with human anatomical heuristics. This logic acts as a critical filter for environmental false positives. If a DNRM marker is detected on a static billboard, a television screen in the background, or a discarded garment lying on a chair, the face-association confidence drops to zero. The protocol discards these non-human detections immediately. This prevents "denial of service" attacks where bad actors attempt to spam or crash the capture pipeline by littering an environment with stickers or markers. The protocol only protects *people*, not objects.

5.4 Obfuscation Timing The critical enforcement requirement is that masking must occur before long-term encoded storage or network transmission. The safest implementation point is inside, or as close as possible to, the camera pipeline buffer. Ideal order of operations:

- frame captured from sensor
- local inference checks for DNRM marker
- face association logic confirms protected subject
- protected facial region isolated
- obfuscation applied to outgoing visible frame

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- raw protected region packaged separately into encrypted compartment
- visible media written with blurred region only

If the raw frame is written to general storage first and blurred later, the privacy promise is weakened.

5.5 Obfuscation Modes The visible output may use one of several methods:

- Gaussian blur
- mosaic pixelation
- opaque fill
- adaptive privacy mask
- adversarial texture scrambling

Different jurisdictions or device policies may prefer different modes. The visible layer must remain sufficient to communicate that the subject was intentionally protected while maintaining scene continuity for non-protected context.

5.6 Multi-Frame Persistence To prevent flicker, mask instability, and momentary reveal windows, DNRM should use temporal persistence logic:

- carry forward previous face estimates
- hold masking for a defined frame timeout after last detection
- interpolate bounding boxes across momentary occlusions
- widen mask slightly under low-confidence motion events

This is especially important for livestreams, rapid movement, and partial camera occlusions.

6. Phase II: Encrypted Biometric Compartment

The protocol must distinguish between the visible obfuscated media and the hidden recoverable biometric segment.

6.1 Protected Region Isolation Instead of encrypting an entire video frame, DNRM isolates only the protected biometric content and its evidentiary context. This dramatically reduces storage burden and limits the recoverable surface area.

The protected package should include:

- facial pixel region
- surrounding context margin
- frame timestamp
- frame index
- device attestation data if available
- marker confidence score
- face association confidence

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- bounding box geometry
- cryptographic hash of the visible frame
- cryptographic hash of neighboring frame references where relevant

This creates a Biometric Evidence Envelope (BEE).

6.2 Envelope Structure Conceptually: BEE = { P, T, F, B, M, Hv, Hd, A }

Where:

- P = protected facial pixel data
- T = timestamp metadata
- F = frame identifier or sequence reference
- B = bounding box and region geometry
- M = marker detection metadata and confidence
- Hv = hash of the visible obfuscated media reference
- Hd = device or capture signature
- A = optional hardware attestation and pipeline provenance

This envelope is then encrypted locally using a unique session key.

6.3 Encryption For each protected event or segment, the device generates a unique symmetric key Ks, ideally using secure hardware randomness. The envelope is encrypted using authenticated encryption such as AES-256-GCM or an equivalent modern AEAD construction.

Cs = Enc_Ks(BEE)

This yields ciphertext Cs, which is the only recoverable form retained alongside the visible obfuscated media.

6.4 Why Segment-Level Encryption Matters This design avoids two bad extremes:

- storing full raw unblurred video
- permanently destroying all evidentiary recoverability

By encrypting only the protected biometric segment, DNRM preserves evidentiary proportionality. Recovery reveals only what was intentionally protected, not the entire original raw stream.

7. Blockchain-Governed Recovery: The Decentralized Recovery and Escrow Network (DREN)

DREN is the core decentralized blockchain layer in V2. Its purpose is not to store video. Its purpose is to prevent unilateral recovery.

7.1 DREN Principles DREN must satisfy all of the following:

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- no single operator can reconstruct recovery keys
- no central company can secretly unlock faces
- no node stores usable footage
- all recovery actions are publicly auditable
- recovery approval is threshold-gated
- mass recovery requests are structurally difficult and visible

7.2 What the Blockchain Stores The blockchain or rollup layer may store:

- encrypted recovery policy references
- warrant request hashes
- recovery request IDs
- node votes or attestations
- approval thresholds
- audit outcomes
- revocations
- protocol versioning
- governance updates
- dispute references

It should not store:

- raw video
- raw facial imagery
- decrypted biometric content
- complete ciphertext payloads where privacy risk or cost is excessive

7.3 What Nodes Hold DREN nodes hold only:

- threshold key shares
- approval logic software
- policy rules
- attestation keys
- node identity credentials
- governance participation rights

Nodes do not hold:

- the full symmetric session key
- raw footage
- reusable master decryption access to all content

7.4 Threshold Cryptography The recovery key material must be split across independent nodes using threshold cryptography. A simple conceptual model is Shamir's Secret Sharing, though production implementations may use threshold signature schemes, distributed key generation, or MPC-backed threshold systems.

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The session key Ks or a wrapped recovery secret is divided into n shares such that at least k shares are required to reconstruct access. Ks -> {s1, s2, ..., sn}

No node alone can recover Ks. Fewer than k nodes reveal nothing useful.

7.5 Key Wrapping Model A practical implementation may use a two-key model:

- The device encrypts the Biometric Evidence Envelope with a session key Ks
- Ks is then wrapped under a recovery key controlled by DREN
- DREN governs only the unwrap step, not the media itself

This is often operationally cleaner than sharding every raw session key directly.

8. Recovery Request Lifecycle

A core strength of DNRM V2 is that recovery is procedural, narrow, and auditable.

8.1 Step 1: Event Creation When a DNRM marker is detected:

- the visible media is obfuscated
- the protected biometric region is isolated
- a Biometric Evidence Envelope is created
- the envelope is encrypted into Cs
- a wrapped recovery path is bound to the encrypted segment
- an evidence manifest is generated

8.2 Step 2: Custody Remains With the Recorder or Hosting System The recorder, platform, or hosting party may retain the visible file plus associated ciphertext envelope. They cannot decrypt the protected biometric content on their own.

8.3 Step 3: Lawful Request Submission If recovery is sought in connection with serious alleged unlawful activity, the requesting authority submits:

- signed legal authorization
- case identifier
- evidence hash references
- requested time range
- requested segment scope
- jurisdictional basis
- reason for necessity and proportionality
- proof of relevance to a specific incident

This request is committed to the blockchain as a hashed recovery petition.

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8.4 Step 4: Node Review and Voting Independent DREN nodes review the request according to protocol and governance policy. They do not need access to raw footage to evaluate whether the request satisfies formal conditions.

Nodes assess:

- legal sufficiency
- jurisdictional validity
- scope limitation
- seriousness threshold
- anti-bulk safeguards
- request duplication
- proportionality
- manifest integrity

Each node records an approval or rejection attestation.

8.5 Step 5: Threshold Approval Only if at least k of n nodes approve does the recovery process proceed.

8.6 Step 6: Key Reconstruction or Threshold Decryption Once threshold approval is reached:

- shares are combined, or
- a threshold unwrapping process is executed, or
- the requesting party receives a decryption token valid only for the approved envelope and scope

8.7 Step 7: Local or Controlled Recovery Recovery should occur only against the approved ciphertext segment and only for the warranted temporal and evidentiary range.

8.8 Step 8: Immutable Audit The network permanently records:

- request hash
- scope requested
- scope granted
- approving nodes
- rejection votes if policy permits
- timestamp
- protocol version
- any disputes or appeal markers

This creates public accountability even when underlying case details remain partially private.

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9. Anti-Abuse Safeguards

A recovery system is only legitimate if abuse resistance is built in by design.

9.1 No Unilateral Access No single recorder, OEM, platform, police agency, or government authority should be able to recover protected faces alone.

9.2 Bounded Recovery Recovery must be limited to:

- specific envelopes
- specific frame ranges
- specific case contexts
- specific warrant references

The system should never support “unlock all protected faces in this video” as a default primitive.

9.3 Anti-Bulk Rate Limits Protocol policy should reject or heavily scrutinize requests that exhibit bulk characteristics:

- many files at once
- wide date ranges
- many unrelated subjects
- weak relevance proofs
- recurring same-agency sweeps

9.4 Public Detection of Mass Requests Even if encrypted or partially abstracted, the public chain should reveal that a recovery request occurred and its scale characteristics. This makes hidden mass-decryption programs difficult.

9.5 Jurisdictional Compartmentalization Nodes may be distributed across legal, civil society, technical, and regional boundaries to reduce coordinated capture. Possible node classes:

- civil liberties organizations
- digital rights groups
- independent forensic institutions
- accredited judicial observers
- academic cryptography labs
- regulated public-interest custodians

9.6 Emergency Override Rejection There should be no secret master key and no silent emergency bypass embedded by OEMs or governments.

9.7 Default Failure Mode If the network is unavailable, divided, censored, or below threshold, recovery fails closed. The protected biometric data remains inaccessible.

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10. Social Media and Third-Party Distribution

One of the weakest points in privacy-preserving media systems is downstream platform processing.

10.1 The Metadata-Stripping Problem Most major platforms:

- re-encode video
- strip metadata
- transform formats
- compress uploads
- discard non-standard sidecar information

A recovery model that relies only on fragile metadata will not survive real-world distribution.

10.2 Dual Persistence Strategy DNRM V2 should support a dual persistence design: A. Sidecar or Container Metadata Where supported, ciphertext manifests and recovery references are stored in structured media metadata or sidecar files. B. Compression-Resilient Embedded Recovery Markers A compact recovery reference may also be embedded into the visible media using robust techniques such as:

- watermarking
- transform-domain embedding
- error-corrected carrier blocks
- visible micro-fiducials where necessary

The goal is not to expose raw ciphertext publicly, but to preserve recoverability linkage even after platform transformation.

10.3 Platform Role Compliant platforms should act only as custodians of:

- obfuscated content
- associated encrypted envelope references
- recovery manifests

They should not hold decryption authority. This allows platforms to avoid becoming custodians of viewable biometric material while still preserving lawful evidentiary pathways.

10.4 Retroactive Cloud Ingestion (Platform-Side Enforcement) Not all media will originate from Tier I (Firmware) or Tier II (Software) compliant devices. A user may capture a video on a legacy, non-compliant camera (Tier III) and upload the raw, unencrypted file to a social media platform like Facebook or TikTok. To maintain the visual privacy standard, compliant platforms must act as the enforcer of last resort. During the upload ingestion pipeline, the platform's servers must scan the raw video for DNRM markers. If a marker is detected, the platform retroactively executes the DNRM protocol:

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1. It applies the obfuscation mask to the public-facing video.
2. It creates the Biometric Evidence Envelope (BEE) and encrypts the raw facial pixels locally on the ingestion server using a generated session key Ks.
3. It links the ciphertext to the DREN network for lawful recovery.
4. Crucially, the platform immediately and permanently deletes the raw, unencrypted upload from its ingestion servers.

In this scenario, the platform "upgrades" the media to DNRM compliance. Even if the device owner failed to respect the physical boundary, the distribution network will enforce it.

10.5 Default Pass-Through and Business Continuity The DNRM protocol is an explicit, opt-in privacy framework; it is not a universal media censor. For a platform to adopt DNRM, it does not need to alter its handling of unmarked media. If a video is ingested and the edge-detection pipeline finds zero DNRM physical markers, the platform's standard encoding, compression, and hosting operations proceed exactly as they do today. The DNRM protocol simply idles. This "Default Pass-Through" guarantees that integrating DNRM will not introduce latency, UX degradation, or algorithmic disruption to the vast majority of a platform's standard content. The protocol only activates when explicit, machine-readable non-consent is detected.

10.6 The Digital Retroactive Claim (Markerless Opt-In) The physical DNRM marker is the primary trigger for automated consent. However, the protocol must also account for subjects filmed without physical gear who discover their biometric data hosted on a compliant platform and wish to revoke visual access. In this scenario, platforms can support a Digital Retroactive Claim.

1. The subject submits a verified "Proof of Identity" claim to the platform regarding a specific piece of media.
2. Upon verification, the platform invokes the DNRM protocol digitally.
3. The platform applies the dynamic obfuscation mask to the subject in the public-facing video.
4. The platform generates the Biometric Evidence Envelope (BEE), encrypts the raw facial pixels locally to create the ciphertext C, and binds it to the DREN smart contract for potential lawful recovery.
5. The platform permanently purges the raw, unencrypted file from its active servers.

By offering a digital invocation of the DNRM protocol, platforms provide a standardized, cryptographically secure mechanism for post-capture privacy, extending the protocol's protection to those who were not physically signaling at the time of capture.

11. Threat Model

The DNRM protocol must be judged against realistic adversaries.

11.1 Rogue Recorder The recorder intentionally wants to reveal a protected face. Mitigation:

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- visible media already obfuscated
- protected segment encrypted
- no local access to recovery key
- no unilateral decryption path Residual risk:
- use of second device
- disabled compliant pipeline
- raw sensor compromise

11.2 Malicious Platform A platform wants to harvest or expose protected biometric content. Mitigation:

- platform never receives plaintext protected face if protocol is followed
- only visible obfuscated media and encrypted compartments are stored
- no unilateral key access Residual risk:
- metadata stripping
- refusal to preserve recovery reference
- non-compliant ingestion pipeline

11.3 Government Mass Surveillance A state actor attempts bulk recovery across many files. Mitigation:

- threshold approval required
- public audit trail
- no central unlock key
- anti-bulk policy rules
- distributed node governance Residual risk:
- legal coercion across many jurisdictions
- node capture
- political pressure

11.4 Node Collusion Enough nodes collude to approve improper recovery. Mitigation:

- diverse node distribution
- transparent chain logging
- slashing or removal governance
- public dispute process
- high reputational cost Residual risk:
- sufficiently broad capture of node set

11.5 OEM Backdoor Risk A manufacturer implements nominal DNRM support but silently preserves raw access internally. Mitigation:

- open-source attestation requirements
- secure audit programs
- reproducible reference implementations
- transparency reporting

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- hardware attestation where possible Residual risk:
- proprietary firmware opacity

11.6 Marker Spoofing An attacker wears fake markers to force blur or misuse recovery pathways. Mitigation:

- signed marker generations in future versions
- anti-spoof classification
- protocol-certified designs
- behavioral heuristics
- optional rotating authenticity features Residual risk:
- casual spoofing may still occur in early versions

12. Governance and Protocol Evolution

DNRM must not be frozen. It must evolve through decentralized governance.

12.1 DNRM Improvement Proposals (DIPs) Protocol changes should be introduced through public proposals across several domains:

- DIP-Vision: Marker design, detection architectures, association logic, anti-spoof systems, low-light performance, edge model optimization.
- DIP-Crypto: Threshold cryptography improvements, MPC hardening, recovery key wrapping design, post-quantum migration planning.
- DIP-Chain: Consensus logic, recovery request formats, node registry rules, audit schema, cross-chain or rollup strategies.
- DIP-Edge: Native SDKs, C++, Rust, Swift, Android, browser integrations, ISP and NPU bindings, hardware attestation.
- DIP-Policy: Serious-offense thresholds, lawful request requirements, jurisdictional standards, oversight frameworks, dispute resolution.

12.2 Node Admission and Removal DREN must define:

- node qualification requirements
- transparency obligations
- identity disclosure standards
- conflict-of-interest rules
- slashing or expulsion procedures
- quorum updates
- emergency replacement processes

12.3 Public Oversight Because privacy and law enforcement interests are both implicated, governance should include public reporting mechanisms:

- annual recovery statistics

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- node voting transparency where lawful
- request type aggregation
- geographic distribution metrics
- appeal outcomes
- rejected bulk request reporting

12.4 Protocol Funding and Network Sustainability Operating a decentralized blockchain network, funding independent node operators, and subsidizing smart contract transaction (gas) fees requires sustainable capital. Relying on venture capital introduces the risk of corporate capture, and charging law enforcement per-warrant introduces perverse financial incentives. To maintain absolute independence, the DREN network is funded directly by the ecosystem it protects. Sales of official DNRM physical markers, apparel, and gear directly subsidize the open-source development, smart contract deployment, and ongoing network maintenance of the protocol. This creates a self-sustaining privacy economy: the physical tools used to signal consent directly fund the digital infrastructure required to enforce it.

13. Legal and Forensic Positioning

A key purpose of V2 is to make DNRM credible to both privacy advocates and lawful institutions.

13.1 Why DNRM Is Not an Evidence Destruction Tool The system does not destroy evidentiary possibility. It delays and decentralizes access. Protected biometric content is:

- preserved in encrypted form
- not accessible by ordinary viewers
- recoverable only through threshold-governed lawful process

13.2 Why DNRM Is Not a Surveillance Tool The system does not grant a standing unlock ability to:

- platforms
- device manufacturers
- app operators
- governments
- law enforcement absent threshold approval

13.3 Why DNRM Is Politically Viable A protocol that permanently deletes all protected pixels is unlikely to be adopted by major OEMs or accepted by lawmakers. A protocol that permits narrowly governed recovery under public audit is much more deployable.

13.4 Evidentiary Integrity The Biometric Evidence Envelope structure improves forensic defensibility by binding:

- protected pixels

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- time
- device provenance
- visible file hash
- detection and association metadata

This helps courts distinguish genuine protected segments from altered or fabricated evidence.

14. Architectural Rationale

14.1 Why Decentralized Blockchain Governance Instead of Central Escrow A centralized escrow service would become:

- a biometric honeypot
- a political target
- a surveillance chokepoint
- a liability magnet
- a single point of failure

Blockchain governance introduces:

- public auditability
- multi-party approvals
- resistance to unilateral tampering
- transparent recovery history
- open protocol evolution

14.2 Why Threshold Access Instead of OEM-Controlled Keys If OEMs hold recovery authority, they become the de facto gatekeepers of privacy and evidence. That recreates the exact trust problem DNRM is intended to solve.

14.3 Why Not Permanent Destruction Because permanent destruction is operationally attractive to privacy maximalists but strategically unviable for broad adoption. The purpose of DNRM is not only to be principled, but to become a standard.

14.4 Why Store the File Outside the Chain Video is too large, too sensitive, and too expensive to route through blockchain systems. The chain should govern authorization, not media storage.

14.5 Why Segment-Level Recovery Instead of Whole-File Recovery Whole-file recovery creates excessive exposure and invites abuse. Segment-level recovery is more proportional, more defensible, and more privacy-preserving.

15. Implementation Roadmap

15.1 Stage 1: Open Detection Standard

- publish marker specification

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- release open-source training dataset
- build baseline detector
- support browser and desktop demos
- prove edge viability in real time

15.2 Stage 2: Encrypted Envelope Prototype

- isolate face regions
- build local envelope encryption
- create manifest format
- demonstrate visible blur plus recoverable encrypted segment

15.3 Stage 3: DREN Testnet

- deploy node registry
- test threshold key management
- build recovery request contract
- simulate lawful request workflow
- publish public audit explorer

15.4 Stage 4: SDK and App Integration

- camera app integrations
- livestream pipeline integrations
- editing software plugins
- mobile SDKs
- platform ingestion tools

15.5 Stage 5: OEM and Institutional Engagement

- publish reference architecture
- invite device manufacturers
- engage privacy NGOs
- engage forensic and judicial stakeholders
- propose visual consent standard adoption

16. Limitations

DNRM must be honest about what it cannot solve. It cannot fully prevent:

- non-compliant hidden cameras
- parallel second-device recording
- firmware-level malicious bypass on hostile hardware
- universal international legal harmonization
- all spoofing in early protocol generations
- guaranteed preservation by hostile social media platforms

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DNRM improves the default condition in compliant ecosystems. It does not eliminate all adversarial capture. That limitation is not a weakness in the protocol. It is a reflection of reality.

17. Future Research Directions

Future versions may explore:

- zero-knowledge proofs for scope-limited recovery
- post-quantum threshold cryptography
- verifiable hardware attestation for compliant capture devices
- decentralized identity for node operators
- privacy-preserving dispute arbitration
- signed anti-counterfeit garment markers
- cross-platform recovery manifests
- selective reveal proofs for courts
- on-chain slashing for abusive node behavior
- jurisdiction-aware policy modules

18. Conclusion

The DNRM Protocol proposes a new default for visual privacy: one in which consent is not assumed, protection is not delayed, and lawful accountability does not require centralized trust.

By combining machine-readable physical consent signals, edge-computed facial obfuscation, encrypted biometric compartmentalization, threshold cryptography, and blockchain-governed recovery, DNRM offers a practical architecture for automated visual consent in a world saturated with cameras.

Its core claim is not that privacy must eliminate accountability, nor that accountability must destroy privacy. Its claim is that both can coexist if access is decentralized, narrow, transparent, and cryptographically governed.

DNRM shifts the balance of power away from platforms, away from silent state access, away from opaque device manufacturers, and toward an open protocol in which visual consent becomes technically enforceable and any exception to that protection becomes publicly auditable.

19. Legal Disclaimer & Open Source Notice

No Guarantee of Absolute Privacy

The Do Not Record Me (DNRM) Protocol is an experimental, open-source technological framework designed to enhance visual consent. While the cryptography and edge-computed obfuscation are built to industry standards, no system can guarantee absolute privacy, security,

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or cryptographic permanence against all adversarial threats, undisclosed hardware backdoors, or future quantum-computing advancements. The DNRM protocol operates optimally within compliant ecosystems (Tier I and Tier II); it cannot physically prevent recording by non-compliant or malicious actors (Tier III).

Limitation of Liability

The authors, developers, and node operators of the DNRM Protocol assume no liability for the capture, leakage, decryption, or misuse of biometric data by third parties. Wearing DNRM physical markers does not grant the wearer legal immunity, nor does it override local, state, or international laws regarding public photography, surveillance, or law enforcement investigations.

Not Legal Advice

The frameworks discussed in this document regarding "lawful recovery," "warrants," and "evidentiary integrity" represent a proposed technological architecture, not legal advice. Platform hosts, Original Equipment Manufacturers (OEMs), and decentralized node operators must independently verify compliance with the legal jurisdictions in which they operate.

Open Source & Intellectual Property

The core algorithmic pipeline, smart contracts, and protocol architecture are intended to be open-source public goods. However, the specific DNRM name, logos, and physical marker designs utilized on official apparel and gear are the proprietary trademarks of Do Not Record Me.

Your Image. Your Rules.

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